

Project Title:

An Insider Threat Seems to Targeting Our Company Based on Scan Information



Background Information

```
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.1.2.200 netmask 255.255.255.0 broadcast 10.1.2.255
    inet6 fe80::20c:29ff:fe68:f3a6 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:68:f3:a6 txqueuelen 1000 (Ethernet)
    RX packets 77 bytes 8524 (8.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 54 bytes 6330 (6.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 19 base 0x2000
```

- Conducted security assessment of an internal server.
- Objective: Identify vulnerabilities and potential insider threat activity
- Focused on reconnaissance detection and log analysis



Web Server Security Assessment Using Nmap and Nikto

```
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.1.2.25 netmask 255.255.255.0 broadcast 10.1.2.255
    inet6 fe80::20c:29ff:fef2:d4af prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:f2:d4:af txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 9 bytes 726 (726.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 4 bytes 240 (240.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4 bytes 240 (240.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```



Nmap Scan Results: Identified Open Ports and Services

```
#nmap -A 10.1.2.200
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-08-31 21:14 EDT
Nmap scan report for 10.1.2.200
Host is up (0.0019s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.9 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   3072 88:40:90:08:99:12:6a:ca:18:1e:da:50:41:f5:a4:92 (RSA)
|   256 af:5c:d4:ee:34:5c:cb:5d:24:9b:25:db:a1:bf:56:82 (ECDSA)
|_  256 a1:bf:dd:3f:51:0b:ed:24:ab:0f:93:c7:f0:e3:08:49 (ED25519)
23/tcp    open  telnet   Linux telnetd
80/tcp    open  http     Apache httpd 2.4.41 ((Ubuntu))
|_ http-server-header: Apache/2.4.41 (Ubuntu)
|_ http-title: Apache2 Ubuntu Default Page: It works
MAC Address: 00:0C:29:68:F3:A6 (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.8
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1   1.88 ms 10.1.2.200

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```



Nikto Scan Results: Identified Web Server Vulnerabilities

```
#nikto -h http://10.1.2.200
Nikto v2.5.0

-----
Target IP:      10.1.2.200
Target Hostname: 10.1.2.200
Target Port:    80
Start Time:     2025-08-31 19:25:16 (GMT-4)
-----

Server: Apache/2.4.41 (Ubuntu)
/: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
/: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
No CGI Directories found (use '-C all' to force check all possible dirs)
/: Server may leak inodes via ETags, header found with file /, inode: 2aa6, size: 6004e95f37190, mtime: gzip. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-200-1418
Apache/2.4.41 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
OPTIONS: Allowed HTTP Methods: POST, OPTIONS, HEAD, GET .
/backup/: Directory indexing found.
/backup/: This might be interesting.
8102 requests: 0 error(s) and 7 item(s) reported on remote host
End Time:      2025-08-31 19:25:28 (GMT-4) (12 seconds)
-----

1 host(s) tested
```



Log Analysis: Detection of Nmap Scanning Activity

```
root@umgc-victim:/home/student# cat /var/log/apache2/access.log | grep nmap
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "GET / HTTP/1.1" 200 11192 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "OPTIONS / HTTP/1.1" 200 181 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "OPTIONS / HTTP/1.1" 200 181 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "GET /nmaplowercheck1757354618 HTTP/1.1" 404 452 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "PROPFIND / HTTP/1.1" 405 520 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "POST / HTTP/1.1" 200 11192 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "GET /robots.txt HTTP/1.1" 404 452 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
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10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "POST /sdk HTTP/1.1" 404 452 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
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10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "GET /.git/HEAD HTTP/1.1" 404 452 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "OPTIONS / HTTP/1.1" 200 181 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
```

Log Analysis: Detection of Nikto Scanning Activity

```
root@umgc-victim:/home/student# cat /var/log/apache2/access.log | grep nikto
10.1.2.25 - - [08/Sep/2025:00:45:26 -0400] "PUT /nikto-test-dAn3e7X8.html HTTP/1.1" 405 551 "-" "Mozilla/5.0 (Windows NT 10.0
; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36"
root@umgc-victim:/home/student#
```



Log Analysis Using Linux Tools (grep, awk, head, tail)

```
root@umgc-victim:/home/student# head /var/log/apache2/access.log
```

```
10.1.2.25 - - [08/Sep/2025:00:45:25 -0400] "GET / HTTP/1.1" 200 11229 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36"
10.1.2.25 - - [08/Sep/2025:00:45:25 -0400] "GET / HTTP/1.1" 200 11228 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36"
10.1.2.25 - - [08/Sep/2025:00:45:25 -0400] "GET / HTTP/1.1" 200 11228 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36"
10.1.2.25 - - [08/Sep/2025:00:45:25 -0400] "GET /hzme0PGo.iso2022-kr HTTP/1.1" 404 488 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36"
10.1.2.25 - - [08/Sep/2025:00:45:25 -0400] "GET /hzme0PGo.log HTTP/1.1" 404 488 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36"
10.1.2.25 - - [08/Sep/2025:00:45:25 -0400] "GET /hzme0PGo.2 HTTP/1.1" 404 488 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36"
10.1.2.25 - - [08/Sep/2025:00:45:25 -0400] "GET /hzme0PGo.tml HTTP/1.1" 404 488 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36"
10.1.2.25 - - [08/Sep/2025:00:45:25 -0400] "GET /hzme0PGo.xml+ HTTP/1.1" 404 488 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36"
10.1.2.25 - - [08/Sep/2025:00:45:25 -0400] "GET /hzme0PGo.dbc HTTP/1.1" 404 488 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36"
10.1.2.25 - - [08/Sep/2025:00:45:25 -0400] "GET /hzme0PGo/ HTTP/1.1" 404 488 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36"
root@umgc-victim:/home/student#
```

```
root@umgc-victim:/home/student# tail /var/log/apache2/access.log | grep nmap
```

```
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "OPTIONS / HTTP/1.1" 200 181 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "GET /favicon.ico HTTP/1.1" 404 452 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "OPTIONS / HTTP/1.1" 200 181 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
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root@umgc-victim:/home/student#
```

```
root@umgc-victim:/home/student# awk '{print}' /var/log/apache2/access.log
```

```
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "GET / HTTP/1.1" 200 11192 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
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10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "OPTIONS / HTTP/1.1" 200 181 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "GET /nmaplowercheck1757354618 HTTP/1.1" 404 452 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "PROPFIND / HTTP/1.1" 405 520 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "POST / HTTP/1.1" 200 11192 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "GET /robots.txt HTTP/1.1" 404 452 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "PROPFIND / HTTP/1.1" 405 520 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "POST /sdk HTTP/1.1" 404 452 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
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10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "GET /.git/HEAD HTTP/1.1" 404 452 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "OPTIONS / HTTP/1.1" 200 181 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "PROPFIND / HTTP/1.1" 405 520 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "GET /evox/about HTTP/1.1" 404 452 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
10.1.2.25 - - [08/Sep/2025:14:03:38 -0400] "ZCNH / HTTP/1.1" 501 496 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
```

Investigation Summary

- Conducted web server security assessment using Nmap and Nikto
- Identified multiple web server vulnerabilities
- Detected reconnaissance activity through log analysis
- Determined moderate to high risk due to exposed services
- Recommended remediation to improve security posture
- Potential impact includes unauthorized access, data exposure, and service exploitation



Key Findings

Issues Detected

- Outdated Apache
- Directory indexing enabled
- Missing security headers
- Insecure services
- Lack of firewall restrictions
- Evidence of reconnaissance activity

Remediation Recommendations

- Patch outdated Apache
- Add HTTP security headers
- Disable directory indexing
- Use secure services
- Implement firewall restrictions
- Enable continuous monitoring and alerting

